COMP411/511: Project Report

Serra Yılmaz Eray Önder Eren Trabzon

1 Introduction

- **Problem Definition:**

  Whole Slide Imaging (WSI) has revolutionized pathology by converting tissue samples into detailed, high-resolution digital images. However, the classification of such digital WSIs presents significant challenges due to their enormous size, lack of pixel-level annotations, and complex tissue structures[1]. In this project, we address the problem of subtyping lung cancer into Lung Squamous Cell Carcinoma (LUSC) and Lung Adenocarcinoma (LUAD) using weakly supervised deep learning techniques. Unlike traditional machine learning approaches that struggle with the scale and complexity of WSIs, our method leverages Multiple Instance Learning (MIL), treating each WSI as a 'bag' of smaller patches or instances, which collectively determine the slide-level label[3]. To address these challenges, we use a Transformer-based MIL model—TransMIL—to incorporate morphological and spatial correlations between patches, aiming for more accurate and interpretable subtyping.

- **Motivation:**

  Lung cancer is one of the leading causes of cancer-related deaths worldwide, making its early and accurate diagnosis vital for effective treatment. Differentiating between LUSC and LUAD is particularly important, as these subtypes differ significantly in treatment protocols and prognoses. Although existing MIL-based approaches have shown promise in WSI classification, many rely on the assumption that patches within a WSI are independent and identically distributed (i.i.d.). This assumption neglects the critical spatial and contextual relationships in pathology, as pathologists consider both the tissue structures and how they relate to each other when diagnosing[7]. Inspired by this approach, we use TransMIL to capture these relationships, combining medicine insights and advanced computer vision techniques to improve accuracy and reliability[6].

- **Contributions:**

  Adopting TransMIL as a Baseline: We implement the TransMIL framework, which leverages self-attention mechanisms to model spatial relationships and morphological patterns in WSIs.

  Experimenting with Feature Extractors: To enhance feature representation, we explore the use of pretrained models such as ResNet50 and UNI Encoder, evaluating their impact on classification performance.

  Comprehensive Evaluation: We benchmark the performance of our model on the TCGA-NSCLC dataset, focusing on accuracy and precision-recall metrics, aiming to meet or exceed the current state-of-the-art benchmark accuracy of 88%.

  Advancing Interpretability: Using attention visualization techniques, we aim to highlight the regions and patterns the model considers important, connecting computational predictions with pathological insights.

2 Related work

- **Starting Point:**

  The foundational work for this project is the TransMIL framework, an approach that applies transformer-based architectures to WSI classification. TransMIL addresses the shortcomings of traditional MIL methods, such as neglecting spatial and morphological correlations between image patches, by leveraging self-attention mechanisms to model these relationships. This method demonstrates significant improvements in accuracy and interpretability over earlier approaches, making it an ideal starting point for our study.

- **Related Work:**

  Numerous methods have been proposed for WSI classification using deep learning and MIL frameworks. One of the pioneering approaches is Attention-based Deep Multiple Instance Learning (ABMIL)[6], which introduced trainable attention weights to improve feature aggregation and interpretability. Building on this, Mahmood et al. developed the CLAM model[5], which clusters instance-level features to identify diagnostically relevant regions within WSIs. Li et al. introduced DSMIL[4], a dual-stream architecture that uses contrastive learning to address challenges in unbalanced datasets, achieving more precise decision boundaries.

  More recently, transformer-based methods have gained momentum. Zhang et al. proposed AMIL-Trans[7], which combines channel attention and self-attention mechanisms to enhance feature discrimination in WSI analysis. TransMIL, in particular, advances this trend by incorporating spatial and morphological correlations directly into its attention mechanisms, leading to improved classification accuracy and interpretability. By adopting TransMIL as a baseline, our work builds on these advancements to further explore and enhance transformer-based methodologies in medical image analysis.

3 Method

- **Overview:**

  Our approach aims to implement the TransMIL model for weakly supervised WSI classification, leveraging its transformer-based architecture to model both morphological and spatial correlations between instances. In addition, we compare the standard ResNet50 feature extractor, pretrained with ImageNet, with the UNI Encoder[2], a Vision Transformer pretrained on medical images. This experiment evaluates their respective impacts on classification performance.

- **Model Details:**

  TransMIL is designed to classify WSIs in pathology diagnosis by effectively leveraging both morphological and spatial information. The model pipeline is organized into two main phases: **input processing** and the **Transformer-based Patch Tokenization (TPT) module**.

  - **Phase 1: Input Processing**
    Feature Embedding Extraction: Each WSI is divided into non-overlapping patches of size 256x256. Background patches are discarded, and the remaining patches are processed using a ResNet50 pre-trained on ImageNet. The ResNet50 outputs 1024-dimensional feature embeddings for each patch, subsequently reduced to 512 dimensions using a fully connected layer.

    Bag Formation: The extracted feature embeddings for all patches in a WSI form a "bag," enabling MIL. Each bag represents an individual WSI.

- **Phase 2: TPT Module**
  Sequence Preparation: The patch embeddings from the input phase are padded or cropped to fit a square sequence. A class token is appended to the sequence, acting as the global representation of the bag.

  Correlation Modeling: Using a Multi-Head Self-Attention (MSA) mechanism, the TPT module models relationships between patch tokens, considering both morphological and spatial correlations.

  Positional Encoding: The Pyramid Position Encoding Generator (PPEG) is introduced to encode spatial information at multiple granularities. The PPEG proposes multi-scale spatial awareness to the sequence by reshaping patch tokens into a 2D grid, applying convolutions with varying kernel sizes to capture spatial relationships, and fusing the outputs to enrich the positional information of each token before re-flattening the sequence for Transformer processing.

  Deep Feature Aggregation: The self-attention outputs are passed through another Transformer layer, allowing for further integration of contextual information across the sequence.

  Classification: The output of the TPT module is processed through a Multi-Layer Perceptron (MLP) applied to the class token. This provides the bag-level prediction.

- **Training Details**:

  During the training of the TransMIL model, the cross-entropy loss function was utilized to optimize the classification tasks. The optimization process was carried out using the Lookahead optimizer, configured with a learning rate of 2×10^{-4} and a weight decay of 1×10^{-5}. The training employed a mini-batch size of 1, enabling efficient handling of the data. To extract features from the image patches, a ResNet50 model was employed, producing a 1024-dimensional feature vector for each patch. These high-dimensional features were subsequently reduced to 512 dimensions using a fully connected layer before being fed into the model. During the inference phase, softmax normalization was applied to the predicted scores for each class. All experiments and model training were conducted using an NVIDIA Tesla V100 GPU, ensuring computational efficiency and scalability for processing large datasets.

- **What's New:**

  We replaced the traditional ResNet50 feature extractor with a UNI Encoder pre-trained on medical imaging data, conducting a comparative analysis of their performance. Replacing ResNet50 with the UNI Encoder in TransMIL marks an important improvement for analyzing histopathology images. ResNet50, used in the original TransMIL model, was pretrained on ImageNet, a dataset made for general image tasks. While ResNet50 is good at extracting local features, it struggles to fully capture the complex patterns in medical images. On the other hand, the UNI Encoder is pretrained on the Mass-100K dataset, which includes 100 million patches from 100,000 whole-slide images across 20 different tissue types. This specific pretraining allows the UNI Encoder to better extract features that are relevant to medical imaging.

  The architecture of the UNI Encoder is another key advantage. It uses a Vision Transformer (ViT) design, which can analyze both global and local relationships in image patches. This is different from ResNet50's convolutional design, which mainly focuses on local features. The UNI Encoder's ability to understand spatial relationships between patches helps it create more detailed and meaningful feature representations. It also uses a self-supervised learning method called DI-

NOv2, which doesn't rely on large amounts of labeled data. This makes the UNI Encoder more flexible and effective in situations where labeled medical images are limited.

Beyond these technical differences, the UNI Encoder performs better in key medical imaging tasks. It works well with fewer training examples, handles images at different resolutions, and adapts to challenging datasets. While ResNet50 is a strong general-purpose model, it doesn't offer the domain-specific training or advanced feature extraction needed for histopathology. By switching to the UNI Encoder, TransMIL gains more accurate and reliable results, making it much better suited for complex clinical applications.

4 Results

- **Datasets and Metrics:**

  For this study, we gathered 201 diagnostic Whole Slide Images (WSIs) from the TCGA-NSCLC dataset. Our dataset consists of 109 LUAD slides from 102 patients and 92 LUSC slides from 90 patients. The dataset provided only slide-level labels, where each WSI was labeled with its corresponding subtype (LUAD or LUSC), but did not include region-level annotations. We used only the open-access WSIs from the dataset and focused on gathering a sufficient number of samples for training, validation, and testing. Initially, we planned to expand the dataset based on the model's performance, but since our results met our expectations, we decided against collecting additional data.

  To evaluate the performance of our model, we used accuracy as well as precision and recall metrics. Accuracy provided a basic assessment of the overall classification performance, while precision and recall offered more detailed insights into the model's ability to identify true positives and avoid false negatives, which is especially crucial in medical diagnostics. The F1-score, as the harmonic mean of precision and recall, was also calculated to provide a balanced view of the model's performance.

- **Evaluation Protocol:**

  To ensure the model was not influenced by irrelevant features during training, we discarded tissues with black spots, shadows, drawings, or low-quality regions. The dataset was divided into training, validation, and test sets using a 65-20-15 split, ensuring sufficient data for training while retaining enough samples for evaluation. Additionally, we employed 5-fold cross-validation to ensure the robustness and generalizability of our results. Tissue segmentation and stain normalization were performed using the model's default settings to focus on diagnostically relevant regions. Each WSI was divided into non-overlapping patches, which were organized into "bags". Slide-level labels were assigned to these bags, with individual patches inheriting the slide's label.

- **Results:**

  The UNI Encoder model achieved superior performance across all metrics compared to ResNet50. The UNI Encoder's average binary accuracy was 0.8738, with an F1 score of 0.8764, precision of 0.9100, and recall of 0.8506. It achieved an average Class 0 (LUSC) accuracy of 0.9012 and Class 1 (LUAD) accuracy of 0.8506. In contrast, the ResNet50 model demonstrated lower performance, with an average binary accuracy of 0.7298, F1 score of 0.7287, precision of 0.7907, and recall of 0.7167. The average Class 0 accuracy for ResNet50 was 0.7474, and the Class 1 accuracy was 0.7167.

  In addition to these metrics, heatmap outputs were generated to visually interpret the model's attention and highlight the most diagnostically relevant regions of the images. The heatmaps helped to verify the model's focus on important tissue areas and explainability. Some representative results

of the generated heatmaps are presented in the figures below.

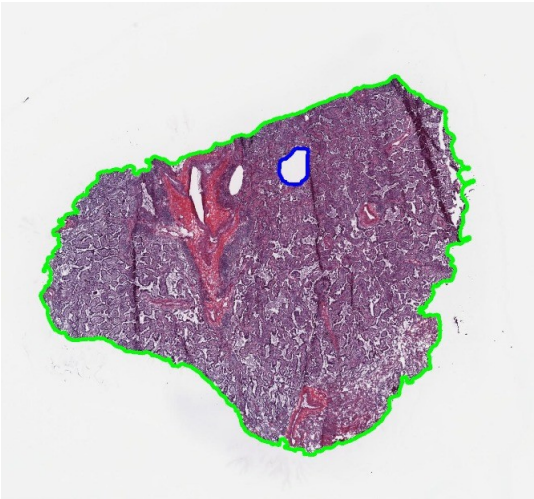

Figure 1: Tissue slide

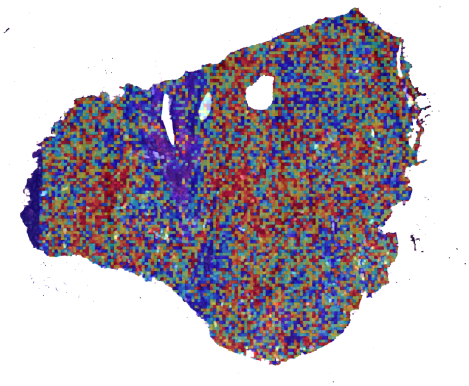

Figure 2: Attention heatmap

Our baseline model, TransMIL, and the CLAM baselines demonstrate that TransMIL consistently outperforms CLAM across the same datasets. TransMIL achieved an accuracy of 0.8835 on TCGA-NSCLC dataset, surpassing CLAM-SB, which achieved an accuracy of 0.8180. These results highlight TransMIL's superior performance compared to CLAM baselines across all three datasets. Notably, we achieved an accuracy of 0.8738 using the UNI Encoder with nearly a quarter of the dataset size used by the authors of the TransMIL method.

- **Ablations and Experiment Conclusions:**

  A series of ablation studies were conducted to assess the importance of various components of our method. Using ResNet50 as the feature extractor achieved an accuracy of 0.72, while replacing it with the UNI Encoder pretrained on medical images improved the accuracy to 0.87.

5 Conclusion

Although we were satisfied with our results, we faced several challenges during the process. Gathering the dataset from the TCGA website was time-consuming, with the downloading process taking much longer than expected. Additionally, we encountered storage issues, which required transferring the dataset to the HPC cluster, which also took significant time. Discarding low-quality images added further delays, as we had to carefully review and discard nearly 50 images. The paper we worked from was published in 2021, which meant we encountered numerous dependency issues, requiring multiple adjustments to the code. Another major challenge was generating heatmaps, as the transformer module uses 8 heads, and each patch has dependencies on its attention scores from other patches. We chose to average the scores over the heads and then took the maximum score for each patch. The heatmap code originally belonged to the CLAM model, which didn't incorporate transformers, so we had to modify the code to correctly extract attention scores from the multiple heads, such as removing unnecessary tokens and paddings.

In conclusion, the proposed method demonstrated competitive performance in LUAD/LUSC classification, achieving state-of-the-art accuracy. The integration of transformer-based attention mechanisms and medical-specific feature extractors was crucial for the success of this approach. These findings not only validate the effectiveness of our method but also contribute valuable insights for future research in computational pathology.

Future work could expand the dataset, test the model on other datasets, explore alternative architectures for better performance, and consult with a medical doctor to validate the correctness of the heatmaps in terms of their clinical relevance.

6 Acknowledgments

Eray Önder : Data collection, data preprocessing, reports
Serra Yılmaz : Model training, reports
Eren Trabzon : Evaluations and results, reports

References

[1] G. Campanella, M. G. Hanna, L. Geneslaw, A. Miraflor, V. Werneck Krauss Silva, K. J. Busam, E. Brogi, V. E. Reuter, D. S. Klimstra, and T. J. Fuchs. Clinical-grade computational pathology using weakly supervised deep learning on whole slide images. *Nature Medicine*, 25(8):1301–1309, Aug 2019. Epub 2019 Jul 15.

[2] R. J. Chen, T. Ding, M. Y. Lu, D. F. K. Williamson, G. Jaume, A. H. Song, B. Chen, A. Zhang, D. Shao, M. Shaban, M. Williams, L. Oldenburg, L. L. Weishaupt, J. J. Wang, A. Vaidya, L. P. Le, G. Gerber, S. Sahai, W. Williams, and F. Mahmood. Towards a general-purpose foundation model for computational pathology. *Nature Medicine*, 30(3):850–862, Mar 2024. Epub 2024 Mar 19. PMID: 38504018; PMCID: PMC11403354.

[3] Maximilian Ilse, Jakub M. Tomczak, and Max Welling. Attention-based deep multiple instance learning. In *Proceedings of the 35th International Conference on Machine Learning (ICML 2018)*, 2018.

[4] Bin Li, Yin Li, and Kevin W. Eliceiri. Dual-stream multiple instance learning network for whole slide image classification with self-supervised contrastive learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR 2021)*, 2021.

[5] Ming Y. Lu, Drew F. K. Williamson, Tiffany Y. Chen, Richard J. Chen, Maria Barbieri, and Faisal Mahmood. Data-efficient and weakly supervised computational pathology on whole-slide images. *Nature Biomedical Engineering*, 5(6):555–570, Jun 2021. Epub 2021 Mar 1.

[6] Zhuchen Shao, Hao Bian, Yang Chen, Yifeng Wang, Jian Zhang, Xiangyang Ji, and Yongbing Zhang. Transmil: Transformer based correlated multiple instance learning for whole slide image classification. In *Proceedings of the 35th Conference on Neural Information Processing Systems (NeurIPS)*. NeurIPS, 2021.

[7] Jun Zhang, Chuang Hou, Wenhao Zhu, Ming Zhang, Yijun Zou, Lucheng Zhang, and Qingchun Zhang. Attention multiple instance learning with transformer aggregation for breast cancer whole slide image classification. In *2022 IEEE International Conference on Bioinformatics and Biomedicine (BIBM)*, pages 1804–1809. IEEE, 2022.